# Language-Gated GR-ConvNet: Steering a Compact Grasp CNN with Frozen CLIP Patch–Text Similarity

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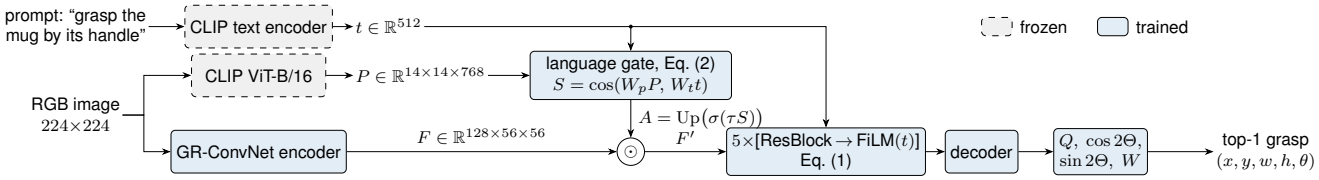



Figure 1. **Language-Gated GR-ConvNet.** Blue: trained (2.9M parameters); CLIP [5]: frozen. The prompt enters through a spatial gate $A$ and through FiLM [4] after every residual block. C0 replaces $W_t t$ by a learned vector $q$.

## Abstract

*Language-driven grasp detection places a grasp on the object part named by a free-form prompt. The proposed method steers the GR-ConvNet grasp CNN with FiLM and a* language gate*: a spatial map of cosine similarities between frozen CLIP patch tokens and the prompt embedding. On Grasp-Anything++, the model reaches 68.6% seen and 57.8% unseen success (two-seed mean) on the official test split, against 54.7% and 40.7% with FiLM alone. A control gate that receives the same CLIP features but not the prompt scores 11.8 unseen points lower in both seeds, so prompt-dependent gating gives 59% of the gain in harmonic mean over FiLM. Code:* https://github.com/Ngoc-LM/language-gated-grasp.

## 1. Introduction

A parallel-jaw grasp [1] is a rotated rectangle $g = (x, y, w, h, \theta)$. Language-driven grasp detection conditions $g$ on a prompt such as "grasp the knife by its handle", so the model must also locate the named part. On Grasp-Anything++ (1M images, 10M prompts), the reference method LGD [8] is a diffusion model. This work tests whether a single-pass grasp CNN can instead be steered by language. The contributions are (i) a *language gate* that re-weights GR-ConvNet [2] features by CLIP [5] patch–text similarity (Fig. 1); (ii) a controlled ablation (text-free gate, two seeds) that separates CLIP visual features from prompt-dependent gating; (iii) an evaluation that excludes test scenes and unseen object names from training, with open code.

## 2. Related Work

Generative grasp CNNs [2, 3] regress dense grasp maps. CLIPort [6] and CROG [7] use CLIP for manipulation and referring grasp synthesis, and LGD [8] denoises grasps under text guidance on data built with foundation models [9]. Pixel–text score maps from CLIP-style encoders drive dense prediction [10]; the language gate uses such a map as a spatial prior inside a grasp CNN.

## 3. Method

**Grasp maps.** As in [2], the network predicts four $224\times224$ maps: quality $Q$, $\cos 2\Theta$, $\sin 2\Theta$ (doubled angle, as a grasp is $180°$-symmetric) and opening $W$. Targets fill the central third of each ground-truth rectangle. The grasp is read at $\arg\max Q$ of the smoothed maps, with $\theta = \frac{1}{2}\,\mathrm{atan2}(\sin 2\Theta, \cos 2\Theta)$ and $h = w/2$.

**FiLM.** The encoder maps the image to $F \in \mathbb{R}^{128\times56\times56}$, followed by five residual blocks and a decoder. With $t$ the normalised CLIP text embedding, each residual block $k$ is followed by

$$F \leftarrow \big(1 + \Delta\gamma_k(t)\big) \odot F + \beta_k(t), \qquad (1)$$

where $\Delta\gamma_k, \beta_k \in \mathbb{R}^{128}$ come from a zero-initialised linear layer, so each FiLM layer is the identity at initialisation.

**Language gate (proposed).** FiLM applies the same channel modulation at every pixel, so it can favour one object among several only indirectly. The language gate therefore compares the $14\times14$ post-LayerNorm patch tokens $P$ of a frozen CLIP ViT-B/16 with the prompt and re-weights the



B (FiLM) ✗ C0 (text-free gate) ✗ C (proposed) ✓

Figure 2. **Unseen prompt** "Fetch candle by its flame" (green: GT, red: top-1): only C succeeds. On "lamp by its stand", C0 succeeds, C fails.

| | Model | Seen | Unseen | $H$ | ms |
|---|---|---|---|---|---|
| A | GR-ConvNet (no text) | 27.3 | 27.8 | 27.5 | 4.3 |
| B | + FiLM($t$) | $54.7_{\pm 2.7}$ | $40.7_{\pm 0.6}$ | $46.7_{\pm 1.4}$ | 12.3 |
| C0 | B + text-free gate | $63.3_{\pm 0.5}$ | $46.0_{\pm 1.9}$ | $53.3_{\pm 1.1}$ | 18.2 |
| C | B + language gate | $\mathbf{68.6}_{\pm 0.5}$ | $\mathbf{57.8}_{\pm 1.9}$ | $\mathbf{62.8}_{\pm 1.3}$ | 18.9 |
| D | C + gate supervision | 70.9 | 57.7 | 63.6 | 18.8 |

Table 1. **Success rate (%) on the annotated test split.** B, C0, C: mean±half-range over two seeds; A, D: one seed. ms: batch-1 end-to-end T4 latency incl. CLIP.

encoder output:

$$S_{ij} = \cos(W_p\, p_{ij},\, W_t\, t), \\ A = \mathrm{Up}_{56}\big(\sigma(\tau S)\big), \qquad F' = F \odot (1 + A), \tag{2}$$

with learned $W_p \in \mathbb{R}^{256\times 768}$, $W_t \in \mathbb{R}^{256\times 512}$ and temperature $\tau$ (initialised to 10). The residual form $1+A$ only amplifies prompt-matching regions. The **C0 control** replaces $W_t t$ with a learned vector $q$: its gate sees the same CLIP features but not the prompt (FiLM still receives $t$). **Loss:** Smooth-$\ell_1$ summed over the four maps. Variant D adds $0.5\,\mathrm{BCE}(\tau S, M)$, $M$ the pooled ground-truth $Q$ mask.

## 4. Experiments

**Protocol.** Training uses 35,964 Grasp-Anything++ scenes (183,559 prompt–grasp pairs) and validation 734. Testing follows the official LGD split (seen: the 70% most frequent LVIS categories), restricted to annotated pairs: 1,376 seen and 601 unseen from 727 of 803 object keys. Test scenes, and training pairs whose object name is an unseen test name, are excluded from training. A top-1 grasp is correct if its IoU exceeds 0.25 and its angle error is below 30° for some ground-truth grasp of the prompted part; $H$ is the harmonic mean of seen and unseen success. Decoding ground-truth maps scores 97.5% on 200 validation pairs, the ceiling of the encoding. All runs use AdamW (lr $10^{-3}$, cosine), batch 32, 6 epochs, no augmentation and one T4 (2.5–2.7 h); checkpoints are selected on 1,000 validation pairs.

**Results.** Each row of Tab. 1 adds one component; A ignores the prompt, so its seen and unseen scores coincide (27.3 vs. 27.8). FiLM (B) raises $H$ by 19.1 points, CLIP features behind the text-free gate (C0) add 6.6, and letting the gate read the prompt (C) adds 9.5. Within each seed, C exceeds C0 by 11.8 unseen points and by 6.3 and 4.4 seen points. These gaps exceed both the seed half-ranges (at most 1.9) and the 95% interval of a two-proportion difference (±5.6 unseen, ±3.5 seen). Prompt-dependent gating thus accounts for 69% of the B→C gain on unseen objects, 59% in $H$ and 38% on seen objects, where CLIP features contribute most (Fig. 2). Gate supervision (D, one seed) stays within 2.3 points of C on every split, inside these intervals, so the simpler C is retained.

**Cost and limitations.** C adds 0.33M trained parameters to B and a frozen 149M-parameter CLIP (12.5 ms per frame with a cached prompt). Training uses 3.6% of the images; the intervals ignore that prompts share scenes. The prompt-blind A already exceeds the 18 unseen that LGD [8] reports for GR-ConvNet with CLIP, so the protocols differ and LGD's 48/42 (seen/unseen) are not compared.

## 5. Conclusion

A frozen-CLIP patch–text gate raises the $H$ of a FiLM-conditioned GR-ConvNet from 46.7 to 62.8; the text-free control attributes 9.5 points to prompt-dependent gating.

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